

Riley Burton

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EDUCATION:

- **University of Massachusetts, Amherst.** PhD Mathematics. (In progress)
Advisor: Kristin DeVleming (UCSD)
- **University of Oregon.** BS Mathematics, Minor in Physics, 2023.
Graduated summa cum laude

EMPLOYMENT:

- **UMass Amherst**, Graduate Teaching Assistant. Fall 2023 - Present.
- **UO Mathematics Library**, Math Tutor. Fall 2020 - Spring 2023.
- **Staccato Gelato**, Assistant Production. Summer 2017 - Summer 2019.

TEACHING EXPERIENCE:

- UMass Amherst:

Instructor of Record:

- Math 131 (Calculus I): Fall 2025.

Teaching Assistant:

- Math 411 (Introduction to Abstract Algebra I): Summer 2026.
- Math 233 (Calculus III): Spring 2026.
- Math 370 (Writing in Mathematics): Summer 2025.
- Math 300 (Fundamental Concepts of Mathematics): Fall 2024, Spring 2025¹, Summer 2025, Summer 2026.
- Math 132 (Calculus II): Spring 2024, Summer 2024.
- Math 131 (Calculus I): Fall 2023.

¹Recipient of the UMass TA award for excellence during AY2024-2025.

TALKS GIVEN:

Graduate:

- *Triangle Singularities and K3 Surfaces*. UMass Algebraic Geometry Learning Seminar, Spring 2026.
- *Lagrangian Floer Cohomology*. UMass Symplectic Geometry and Fukaya Category Learning Seminar, Fall 2025.
- *What are the Adèles?* UMass Graduate Student Number Theory Seminar, Spring 2024.
- *Dirichlet's Theorem on Primes in Arithmetic Progressions*. UMass Graduate Student Number Theory Seminar, Spring 2024.
- *What is a Modular Form?* UMass Graduate Student Number Theory Seminar, Fall 2023.

Undergraduate:

- *The Algebraic and Geometric Group Laws on an Elliptic Curve*. UO Directed Reading Program, Spring 2023.
- *Introduction to Algebraic Number Theory*. Northwest Undergraduate Mathematics Symposium (NUMS), Fall 2022.
- *Geometry of Numbers and the Finiteness of the Class Group*. UO Directed Reading Program, Spring 2022.
- *Algebraic Number Theory and Fermat's Last Theorem*. UO Directed Reading Program, Spring 2021.
- *Pell's Equation*. UO Directed Reading Program, Spring 2020

PROJECTS/WRITINGS:

Graduate:

- *A Wall Crossing Square for the Hassett-Keel Program in Genus Two*. Fall 2025.

Undergraduate:

- *Algebraic Number Theory*. UO Undergraduate Summer Mathematics Research Program, Summer 2022.
- *Binary Quadratic Forms and the Class Group*. UO Undergraduate Summer Mathematics Research Program, Summer 2021.

CONFERENCES/WORKSHOPS ATTENDED:

Graduate:

- *Computations on K3 Surfaces and Related Varieties*. ICERM (Brown University), September 9 - December 11, 2026 (Upcoming).
- *Summer School on Moduli of Varieties*. SLMath, July 20 - 31, 2026 (Upcoming).
- *Forty Years of Higher Dimensional Geometry*. Princeton University, June 8 - 12, 2026 (Upcoming).
- *AGNES (Algebraic Geometry Northeastern Series)*. Rutgers University, May 1 - 3, 2026.
- *Boston Algebraic Geometry Day*. Harvard University, March 7 2026.
- *AGNES (Algebraic Geometry Northeastern Series)*. UMass Amherst, November 21 - 23, 2025.
- *2025 Summer Research Institute in Algebraic Geometry*. Colorado State University, July 14 - August 1 2025.

Undergraduate:

- *NUMS (Northwest Undergraduate Mathematics Symposium)*. Portland State University, November 12, 2022.

AWARDS/SCHOLARSHIPS:

Graduate:

- UMass TA/Discussion Leader Award for Excellence

Undergraduate:

- Hearn Family Scholarship
- Trigueiro Math Scholarship
- Clarence and Lucille Dunbar Scholarship
- Mercer Family Foundation Scholarship
- Erwin and Gertrude Julifs Scholarship in Mathematics

OTHER SKILLS:

- $\text{\LaTeX}$
- Python
- Adobe Photoshop/Illustrator